Core Finance

MGMP 543

Prof. Benedict Guttman-Kenney

Spring 2025

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Jones Graduate School of Business

Welcome!

My Background

- From England
- Worked in financial services regulation for six years
- University of Chicago Booth School of Business
  - PhD Economics, MBA
- Rice Finance
- My research is focused on credit cards, credit information, buy now pay later, payday loans.

Who Are You?

Course Logistics

What Do You Want From This Course?
In Your Words…

What Should You Get From My Course?
My Aims…

Course Outline: Weeks 1 to 6

Week	Topics	Textbook Chapters (Ross, Westerfield, Jaffe, & Jordan)
1 (17, 19 February)	Time Value of Money	4
2 (24, 26 February)	Capital Budgeting Tools, Bonds	5, 8
3 (3, 5 March)	Equities	9
4 (10, 12 March)	Statistics, Risk and Return, Portfolio Theory	11
5 (17, 19 March)	CAPM	10, 11
6 (24, 26 March)	Beta Management (Case), Course Review Midterm released after your section’s class finishes. Exam must be completed by: 6:00pm Monday 31 March for Section 001 / 6:00pm Wednesday 2 April for Section 002.	-

Course Outline: Weeks 7 to 12

Week	Topics	Textbook Chapters
7 **(31 March, 2 April)**	Efficient Markets, Behavioral Finance	14
8 **(7, 9 April)**	Financing and Capital Structure, MM Theory	16, 17
9 **(14, 16 April)**	Boeing 7E7 (Case), Discounted & Free Cash Flows	-
10 **(21, 23 April)**	Options	22
11 **(28, 30 April)**	Consumer Finance, Current Topics (ESG?, FinTech?, Great Recession? Bank Runs? CFPB?)	-
12 **(5, 7 May)**	Course Review Final released after your section's class finishes. Exam must be completed by: 10pm Monday 12 May for Section 001 / 10pm Wednesday 14 May for Section 002.	-

Graded Assignments

- All assignments are to be completed individually, except the two cases that are in groups.
- **Exams:** Online.
- **Cases:** Two cases to write-up and discuss in class.
- **Quizzes**: Short way to recap class material.
- **Participation:** Expect cold calling in class.
- **Canvas Discussion Board:** Share relevant articles, ask / answer questions!
- **Artificial Intelligence (AI):** The abilities of AI tools such as ChatGPT are rapidly developing. AI tools can assist learning, but they also commonly make basic mistakes. You may use AI tools if it helps you to review concepts learnt in class. AI tools must not be used for any graded assignments (e.g., quizzes, cases, midterm or final exams).

Graded Assignments	Weight
Midterm Exam	30%
Final Exam	30%
Case Write-up	15%
Quizzes	10%
Participation	15%
Total	**100%**

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Teaching Team

Reese Alexander

Luis Levya

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Any Questions?

Week 1:
The Time Value of Money

Today's Topics

1. Opportunity Cost
2. Future Values (FV) and Present Values (PV) of Cash Flows
3. Perpetuities and Annuities

Opportunity Cost

What is the value of a water well?

A) A water well that has run dry is worth…

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What is the value of a water well?

A) A water well that has run dry is worth…

B) A water well connected to an active aquifer is worth…

What is the value of a water well?

A) A water well that has run dry

B) A water well connected to an active aquifer

Value of these assets ***differ***.

- An asset's value is the stream of value it will create in the future.
- Valuation is different to backwards-looking accounting "book value".

What is a one-dollar bill worth?

- Today?

What is a one-dollar bill worth?

- Today?

- One Year From Now?

When would you prefer to receive $1?

a) 2 Years from Now
b) 1 Year from Now
c) Next Week
d) Today

Why?

Opportunity Cost

Money today worth more than in the same amount of money the future due to its earnings capacity.

- Save and earn interest / invest and earn returns
- Pay down debt and reduce interest
- Spend and enjoy now

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“Opportunity Cost”
– choosing one option comes at the cost of not being able to do the next best option.

The Time Value of Money: $1 Now > $1 Later

Future Values (FV) of Cash Flows

Suppose you lend the US government $100 at an interest rate of 20% per year? How much will you have in one year?

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$$FV_1 = PV_0 \times (1 + r)$$

Calculating Future Value

$$FV = PV \times (1 + r)^t$$

FV is "Future Value" (what you are trying to work out in example)

PV is "Present Value" ($100 in example)

r is "Rate of Return" (20% in example)

t is time periods

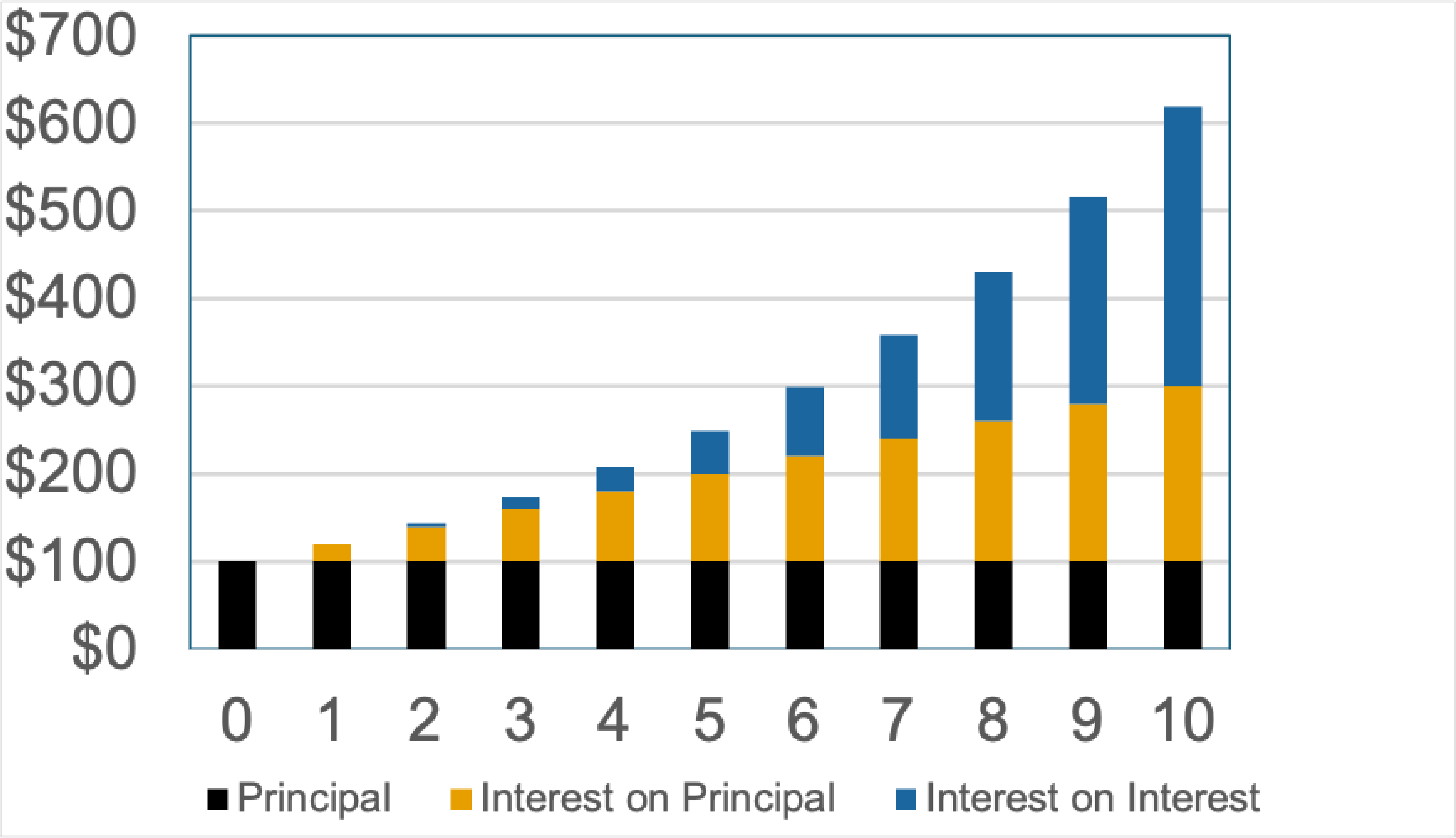
Compounding (Interest Earning Interest)

Suppose you lend the US government $100 at an interest rate of 20% per year. How much will you have **in four years if you re-invest interest each year**?

$$FV_t = PV_0 \times (1 + r)^t$$

Year	Amount
0	**$100**
1	
2	
3	
4	

Effects of Compounding (Interest Earning Interest)



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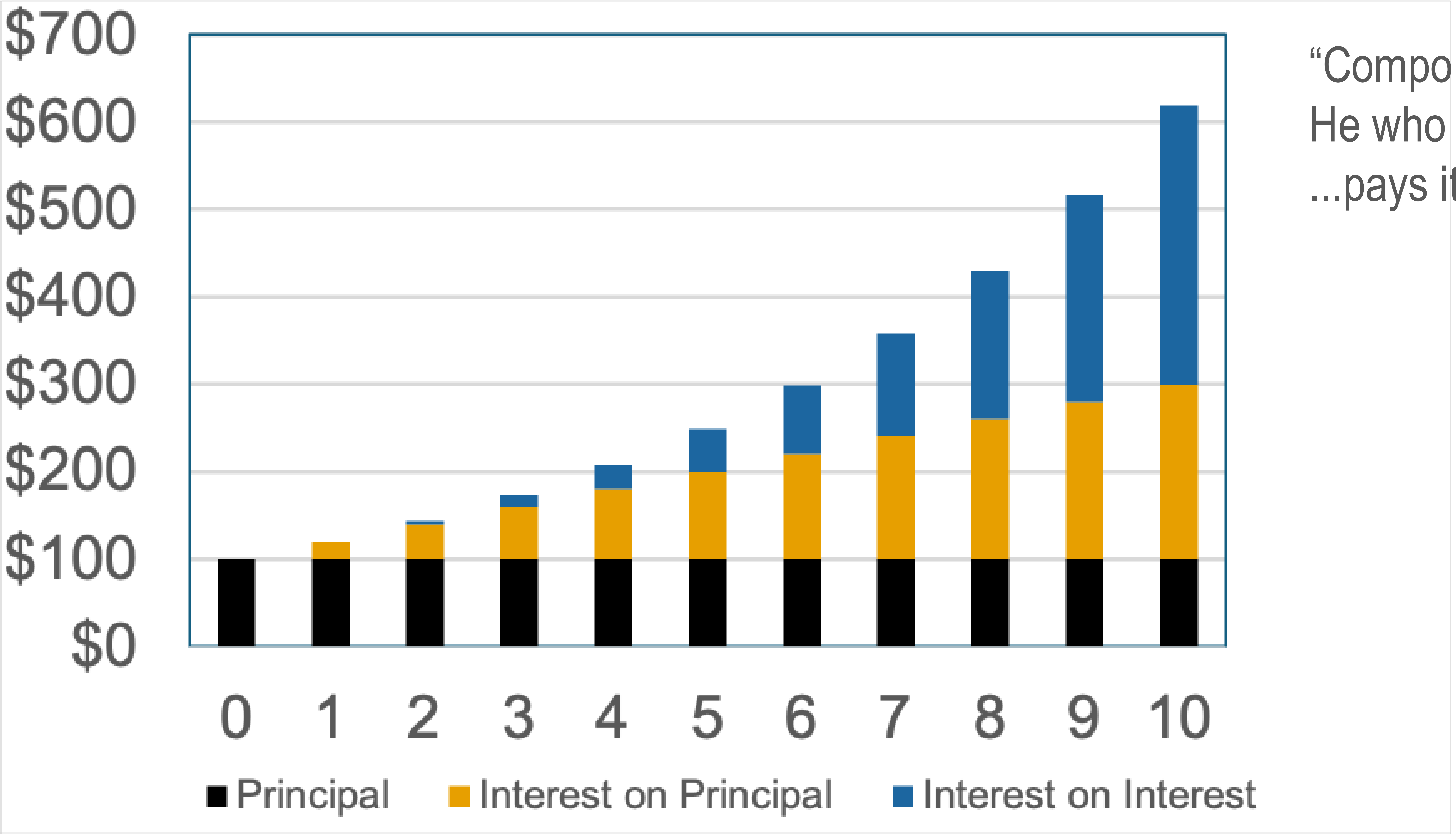
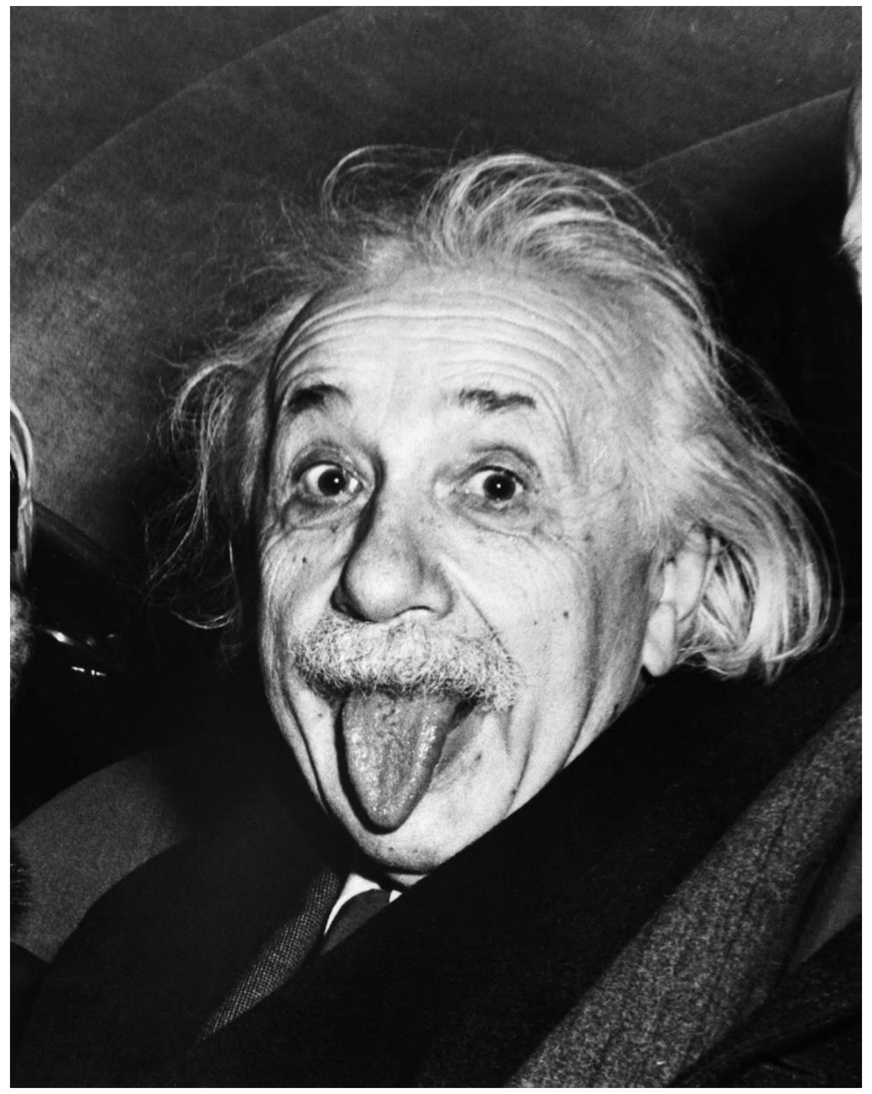
Effects of Compounding (Interest Earning Interest)



"Compound interest is the 8th wonder of the world. He who understands it, earns it...he who doesn't ...pays it." (Einstein)



Present Values (PV) of Cash Flows

Calculating Present Value (PV)

$$FV = PV \times (1 + r)^t$$

Present Value (PV) of Cash Flows

$$PV = \frac{FV}{(1+r)^t}$$

What happens to PV if r ⬆️?

Present Value (PV) of Cash Flows

$$PV = \frac{FV}{(1+r)^t}$$

What happens to PV if r ⬆?

What happens to PV if t ⬆?

Present Value (PV) with Multiple Cash Flows

Cash Flow C_0 today, C_1 at end of first year 1, C_2 at end of year 2, etc

Present Value (PV) with Multiple Cash Flows

$$PV = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

Interest Rates

Who do you think you r?

$$PV = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

r is the:

- Interest Rate
- Discount Rate
- Hurdle Rate (HR / Minimum Acceptable Rate of Return / MARR)
- Required Return
- Cost of Equity (COE / Ke)
- Cost of Debt (COD / Kd)
- Weighted Average Cost of Capital (WACC / Cost of Capital)

Nominal Interest Rate or Effective Interest Rate?

- **Nominal interest rate.** For example, 12% annual interest rate. You pay 1% interest every month but that doesn't consider compounding.

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Discrete time (1,2,3, etc)

$$r_{effective} = (1 + \frac{r_{nominal}}{m})^{m} - 1$$

Continuous time

$$r_{effective} = e^{r_{nominal}} - 1$$

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Adjusting Interest Rate to Align With Cash Flow Timing

$$r_{effective} = (1 + \frac{r_{nominal}}{m})^{m} - 1$$

How to calculate rate if cash comes each month?

Adjusting Interest Rate to Align With Cash Flow Timing

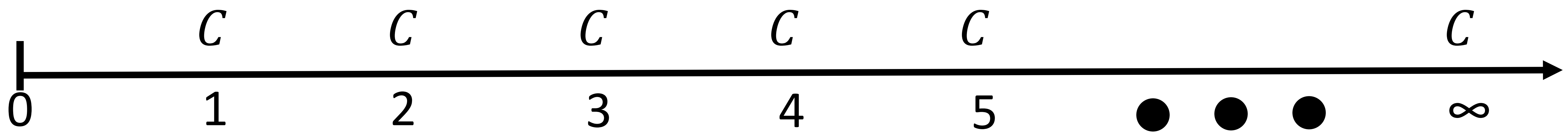
$$r_{effective} = (1 + \frac{r_{nominal}}{m})^{m} - 1$$

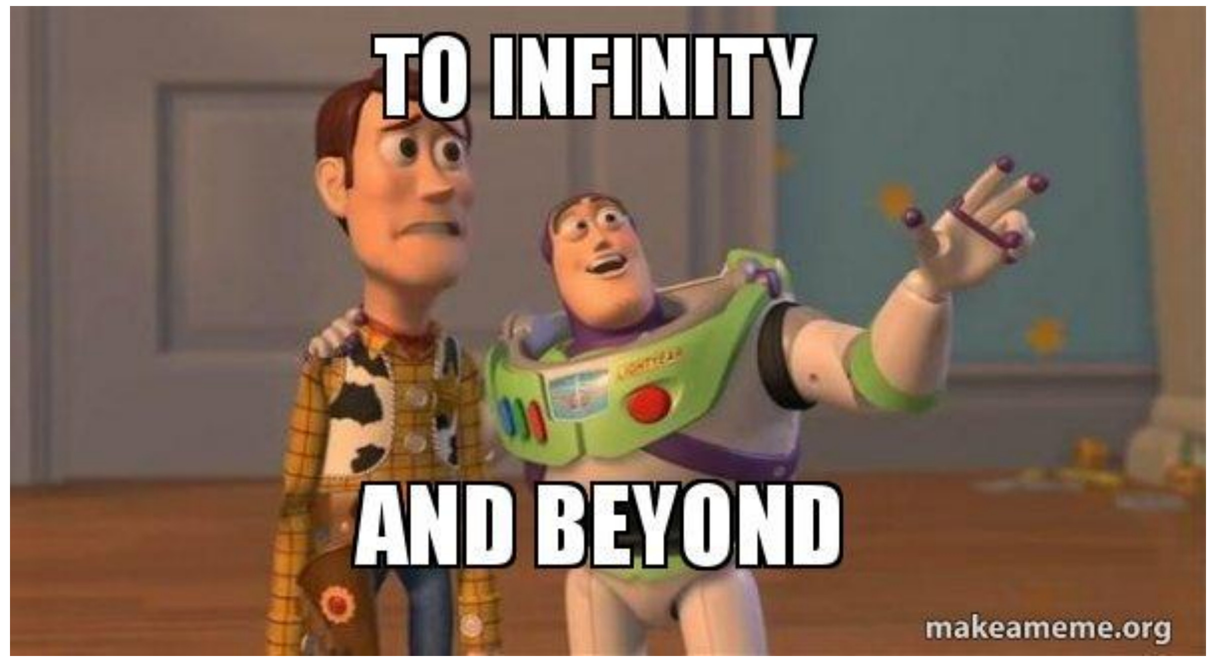
How to calculate rate if cash comes each month?

How to calculate rate if cash comes every 2 years?

Perpetuities

What is a **Perpetuity**?





Why do we care about valuing perpetuities?

Good approximation for things expected to last a long-time (even if not forever):

- Perpetual Bonds (e.g., Consol Bonds)
- University Endowments: Rice University has $7.5 bn Endowment Assets Under Management (AUM) supports 40% of University expenses, tuition discounts.
- Stocks (week 3)

Will be a useful mathematical tool for valuing.

What is the present value of a perpetuity?

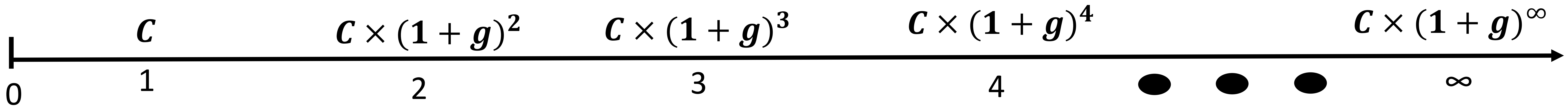
Perpetuity produces a

constant cash flow (C) in every ***future*** period (t = 1,2,…,∞)

$$\boldsymbol{PV = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t} =}$$

What is a **Growing Perpetuity**?

Growing perpetuity produces a cash flow (C)
that **grows at rate g** in every ***future*** period ($t = 1,2,\ldots,\infty$)



What is the present value of a **Growing Perpetuity**?

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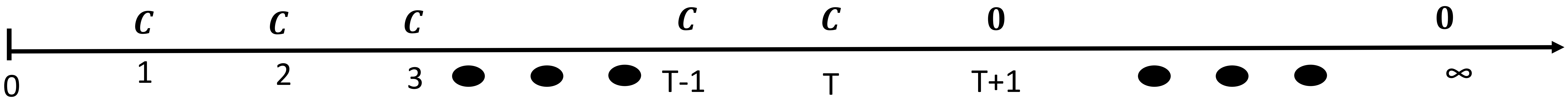
that **grows at rate g** in every ***future*** period ($t = 1,2,\ldots,\infty$)

$$PV = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t} =$$

Annuities

What is an **Annuity**?



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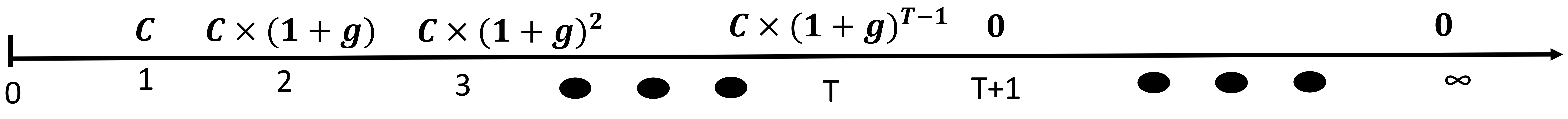
What is the present value of an **Annuity**?

Annuity produces a constant cash flow (C_1)

For a finite number of *future* periods ($t = 1,2,\ldots,T$)

$$\boldsymbol{PV} = \sum_{\boldsymbol{t}=\boldsymbol{0}}^{\infty} \frac{\boldsymbol{C_t}}{(\boldsymbol{1}+\boldsymbol{r})^{\boldsymbol{t}}} =$$

What is a **Growing Annuity**?



What is the present value of a **Growing Annuity**?

Growing annuity produces a cash flow (C)

For a finite number of future periods (t=1,2,…,T)

that **grows at rate g** in every ***future*** period ($t = 1,2,\ldots,\infty$)

$$\boldsymbol{PV = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t} =}$$

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Examples

1. How much money would you have in one year if you lend $1,000 today at an interest rate of 10%?

a) $1,000
b) $1,100
c) $1,150
d) Something else

2. What is the future value of investing $1,000 for 3 years at 10%?

a) $1,100
b) $1,210
c) $1,300
d) $1,321
e) $1,331
f) $1,333
g) Something else

3. What is the present value of $1,000 that will be paid to you in 10 years from now? The discount rate is 10%.

a) $10
b) $100
c) $1,000
d) $10,000
e) Something else

4. You’ve won a prize at the Houston Rodeo and have a choice. If the interest rate is 10%, which is the more valuable prize? (i) $1,000 now or (ii) $1,500 at the end of four years.

a) $1,000 now
b) $1,500 in four years
c) They have the same value
d) Not enough information

5. You've won cash-forever lottery that will pay you and future generations of your family $2,000 every year, starting today. If the interest rate is 10%, What would you value more?

a) $2,000 every year
b) $22,000 now, with no future payments
c) They have the same value
d) Not enough information

6. You’ve won cash-forever lottery that will pay you and future generations of your family every year. The first payment is $3,000 starting one year from today and grows by 2% each year. If the interest rate is 10%, What would you value more?

a) $3,000 next year, with payments growing by 2% each year
b) $35,000 now, with no future payments
c) They have the same value
d) Not enough information

7. You receive $500 every year for the next 20 years, starting in one year. What is the present value at a discount rate of 10%?

a) $10,000
b) $5,000
c) $4,257
d) $3,268
e) $743
f) Not enough information

8. You receive $500 every year for the next 20 years, starting in one year but growing at 5% a year.
What is the present value at a discount rate of 10%?

a) $6,257
b) $6,056
c) $5,000
d) $4,257
e) $3,268
f) Not enough information